# CS2 Dataset Overview and Exploratory Analysis

**Round-level, round-state, and enriched data**

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# 1 Setup

This report combines the dataset overview, round-level descriptive analysis, team/map/side comparisons, round-state analysis, and bomb/economy/player-state analyses into one logically ordered PDF report. Summary tables appear in the report and are also exported as CSV files to `../data/analysis`. Figures are saved as PDF files under `../report/figures/cs2-combined_analysis`; CSV tables are saved under `../data/analysis/cs2-combined_analysis`. These paths assume this QMD is stored in `cs2-rop/R`.

```
library(tidyverse)
library(arrow)
library(knitr)
library(scales)

rounds_file <- "../data/analysis/rounds.parquet"
states_file <- "../data/analysis/round_states.parquet"
enriched_file <- "../data/analysis/round_states_enriched.parquet"
figures_dir <- "../report/figures/cs2-combined_analysis"
tables_dir <- "../data/analysis/cs2-combined_analysis"
dir.create(figures_dir, showWarnings = FALSE, recursive = TRUE)
dir.create(tables_dir, showWarnings = FALSE, recursive = TRUE)

save_csv <- function(data, filename) {
  readr::write_csv(data, file.path(tables_dir, filename))
}

rounds <- read_parquet(rounds_file) |> as_tibble()
round_states <- read_parquet(states_file) |> as_tibble()
round_states_enriched <- read_parquet(enriched_file) |> as_tibble()

map_label <- function(x) tools::toTitleCase(sub("^de_", "", as.character(x)))
rounds <- rounds |> mutate(map = map_label(map))
round_states <- round_states |> mutate(map = map_label(map))
round_states_enriched <- round_states_enriched |> mutate(map = map_label(map))
```

```r
show_table <- function(x, digits = 2) {
  # Use LaTeX tables for PDF output. Long tables can continue across pages.
  knitr::kable(
    x, digits = digits, format = "latex", booktabs = TRUE,
    longtable = nrow(x) > 25, row.names = FALSE
  )
}
save_pdf <- function(plot, filename, width = 9, height = 6) {
  ggsave(file.path(figures_dir, filename), plot = plot,
         width = width, height = height, device = "pdf")
}
theme_set(theme_minimal(base_size = 11))
pct <- scales::label_percent(accuracy = 1, scale = 1)
```

# 2 1. Dataset structure and quality

## 2.1 1.1 Overview of the datasets

`rounds` has one record per round; `round_states` has repeated snapshots over time within a round. `round_states_enriched` attaches selected round-level fields to every snapshot. It is a merged, deduplicated-by-column-selection analytical table: it keeps the snapshot rows and adds round context, making it useful for examining how round-level outcomes relate to the evolving state of a round. It is not one row per round.

```r
dataset_overview <- tibble(
  dataset = c("rounds", "round_states", "round_states_enriched"),
  rows = c(nrow(rounds), nrow(round_states), nrow(round_states_enriched)),
  columns = c(ncol(rounds), ncol(round_states), ncol(round_states_enriched)),
  missing_cells = c(sum(is.na(rounds)), sum(is.na(round_states)),
                    sum(is.na(round_states_enriched))),
  unique_matches = c(n_distinct(rounds$match_id),
                     n_distinct(round_states$match_id),
                     n_distinct(round_states_enriched$match_id)),
  unique_rounds = c(
    nrow(distinct(rounds, match_id, map, round_num)),
    nrow(distinct(round_states, match_id, map, round_num)),
    nrow(distinct(round_states_enriched, match_id, map, round_num))
  )
)
show_table(dataset_overview)
```

| dataset | rows | columns | missing_cells | unique_matches | unique_rounds |
|---|---|---|---|---|---|
| rounds | 1257 | 26 | 1288 | 60 | 1257 |
| round_states | 116201 | 45 | 195710 | 60 | 1257 |
| round_states_enriched | 116201 | 59 | 250917 | 60 | 1257 |

```
save_csv(dataset_overview, "qmd_dataset_overview.csv")
```

## 2.2 1.2 Column names and data types

Only the two original datasets are listed here. If a column appears in both datasets, its `dataset` entry says **Both**. The enriched dataset is derived from these two tables and is described above rather than listing a third, largely repetitive column inventory.

```
round_cols <- tibble(
  column = names(rounds),
  rounds_type = map_chr(rounds, ~ paste(class(.x), collapse = "/"))
)
state_cols <- tibble(
  column = names(round_states),
  round_states_type = map_chr(round_states, ~ paste(class(.x), collapse = "/"))
)
column_overview <- full_join(round_cols, state_cols, by = "column") |>
  mutate(
    dataset = case_when(
      !is.na(rounds_type) & !is.na(round_states_type) ~ "Both rounds and round_states",
      !is.na(rounds_type) ~ "rounds",
      TRUE ~ "round_states"
    ),
    data_type = coalesce(rounds_type, round_states_type)
  ) |>
  select(column, dataset, data_type) |>
  arrange(dataset, column)
show_table(column_overview)
```

| column | dataset | data_type |
|---|---|---|
| bomb_planted | Both rounds and round_states | logical |
| ct_score_before | Both rounds and round_states | integer |
| ct_team | Both rounds and round_states | character |
| ct_win | Both rounds and round_states | logical |
| map | Both rounds and round_states | character |

| | | |
|---|---|---|
| match_id | Both rounds and round_states | character |
| part | Both rounds and round_states | integer |
| plant_site | Both rounds and round_states | character |
| round_num | Both rounds and round_states | integer |
| t_score_before | Both rounds and round_states | integer |
| t_team | Both rounds and round_states | character |
| winner_side | Both rounds and round_states | character |
| bomb_carried | round_states | logical |
| ct_alive | round_states | integer |
| ct_armor | round_states | integer |
| ct_awps | round_states | integer |
| ct_defusers | round_states | integer |
| ct_equipment | round_states | integer |
| ct_flashes | round_states | integer |
| ct_he | round_states | integer |
| ct_helmets | round_states | integer |
| ct_hp | round_states | integer |
| ct_molotovs | round_states | integer |
| ct_money | round_states | integer |
| ct_rifles | round_states | integer |
| ct_smgs | round_states | integer |
| ct_smokes | round_states | integer |
| seconds | round_states | numeric |
| seconds_since_plant | round_states | numeric |
| t_alive | round_states | integer |
| t_armor | round_states | integer |
| t_awps | round_states | integer |
| t_defusers | round_states | integer |
| t_equipment | round_states | integer |
| t_flashes | round_states | integer |
| t_he | round_states | integer |
| t_helmets | round_states | integer |
| t_hp | round_states | integer |
| t_molotovs | round_states | integer |
| t_money | round_states | integer |
| t_rifles | round_states | integer |
| t_smgs | round_states | integer |
| t_smokes | round_states | integer |
| tick | round_states | integer |
| time_remaining_s | round_states | numeric |

| | | |
|---|---|---|
| ct_buy_type | rounds | character |
| ct_equipment_freeze_end | rounds | integer |
| ct_won_previous_round | rounds | logical |
| end_tick | rounds | integer |
| first_death_side | rounds | character |
| freeze_end_tick | rounds | integer |
| is_overtime | rounds | logical |
| is_pistol_round | rounds | logical |
| plant_tick | rounds | integer |
| reason_name | rounds | character |
| round_duration_s | rounds | numeric |
| t_buy_type | rounds | character |
| t_equipment_freeze_end | rounds | integer |
| winner_team | rounds | character |

```
save_csv(column_overview, "qmd_column_overview.csv")
```

## 2.3 1.3 Columns with missing values

Only columns that contain at least one missing value are shown. Missingness is not automatically an error: for example, planting-related fields are naturally unavailable in rounds where the bomb was not planted.

```
missing_by_dataset <- function(data, dataset_name) {
  tibble(
    dataset = dataset_name,
    column = names(data),
    missing_n = map_int(data, ~ sum(is.na(.x))),
    missing_pct = map_dbl(data, ~ 100 * mean(is.na(.x)))
  ) |>
    filter(missing_n > 0) |>
    arrange(desc(missing_n))
}
missing_original <- bind_rows(
  missing_by_dataset(rounds, "rounds"),
  missing_by_dataset(round_states, "round_states")
) |>
  filter(missing_n > 0)
show_table(missing_original)
```

| dataset | column | missing_n | missing_pct |
|---|---|---|---|
| rounds | plant_site | 596 | 47.41 |
| rounds | plant_tick | 596 | 47.41 |
| rounds | ct_won_previous_round | 60 | 4.77 |
| rounds | reason_name | 12 | 0.95 |
| rounds | ct_buy_type | 11 | 0.88 |
| rounds | t_buy_type | 11 | 0.88 |
| rounds | first_death_side | 2 | 0.16 |
| round_states | seconds_since_plant | 97855 | 84.21 |
| round_states | plant_site | 97855 | 84.21 |

```
save_csv(missing_original, "qmd_missing_values_original_datasets.csv")
```

## 2.4 1.4 Maps, teams, and dataset coverage

```
map_overview <- rounds |>
  group_by(map) |>
  summarise(matches = n_distinct(match_id), rounds = n(), .groups = "drop")
show_table(map_overview)
```

| map | matches | rounds |
|---|---|---|
| Dust2 | 30 | 628 |
| Mirage | 30 | 629 |

```
save_csv(map_overview, "qmd_map_overview.csv")

team_overview <- bind_rows(
  rounds |> transmute(team = ct_team),
  rounds |> transmute(team = t_team)
) |>
  filter(!is.na(team), team != "") |>
  distinct(team) |>
  arrange(team)
show_table(team_overview)
```

| team |
|---|
| 9z |
| Aurora |
| FURIA |
| FUT |
| FaZe |
| Falcons |
| G2 |
| Legacy |
| MOUZ |
| NAVI |
| Spirit |
| Vitality |

```
save_csv(team_overview, "qmd_team_overview.csv")
```

# 3 2. Round-level analysis (rounds)

## 3.1 2.1 Round keys and match length

```
round_key_check <- rounds |>
  count(match_id, map, round_num, name = "rows_per_round") |>
  filter(rows_per_round > 1)
tibble(duplicate_round_keys = nrow(round_key_check)) |> show_table()
```

| duplicate_round_keys |
|---|
| 0 |

```
if (nrow(round_key_check) > 0) show_table(round_key_check)

rounds_per_match <- rounds |>
  distinct(match_id, map, round_num) |>
  count(match_id, map, name = "rounds_in_match")
show_table(rounds_per_match |> summarise(
  matches = n(), min_rounds = min(rounds_in_match),
  q1_rounds = quantile(rounds_in_match, .25),
  median_rounds = median(rounds_in_match),
```

```
  mean_rounds = mean(rounds_in_match),
  q3_rounds = quantile(rounds_in_match, .75),
  max_rounds = max(rounds_in_match)
))
```

| matches | min_rounds | q1_rounds | median_rounds | mean_rounds | q3_rounds | max_rounds |
|---|---|---|---|---|---|---|
| 60 | 14 | 18.75 | 21 | 20.95 | 22.25 | 35 |

### 3.2 2.2 Round summary by map

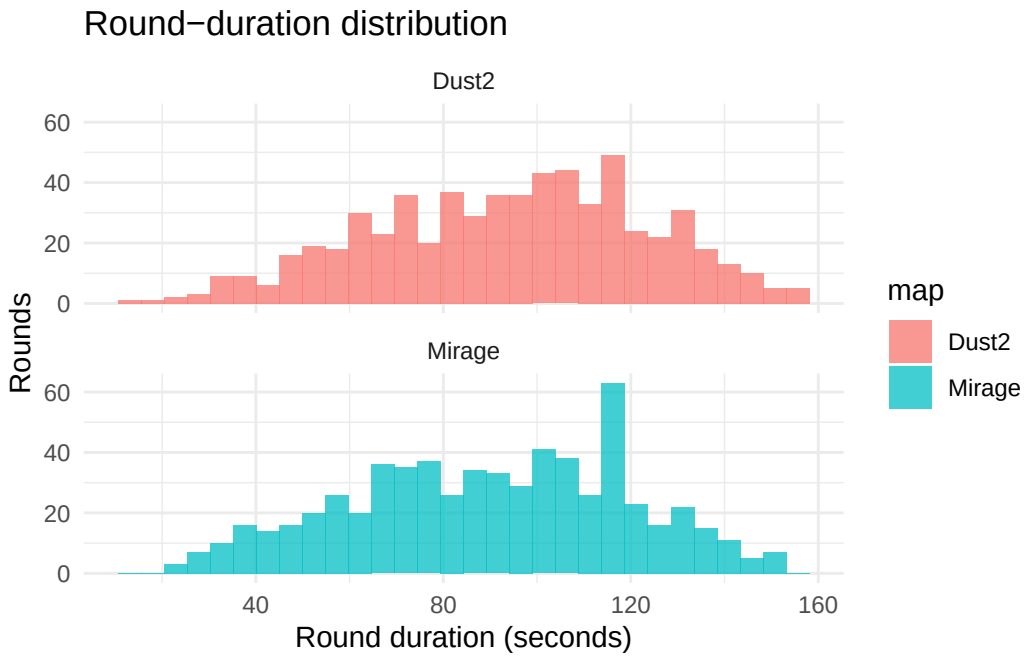
```
rounds_by_map <- rounds |>
  group_by(map) |>
  summarise(
    matches = n_distinct(match_id),
    rounds = n(),
    pistol_rounds = sum(is_pistol_round %in% TRUE, na.rm = TRUE),
    overtime_rounds = sum(is_overtime %in% TRUE, na.rm = TRUE),
    planted_rounds = sum(bomb_planted %in% TRUE, na.rm = TRUE),
    planted_pct = 100 * mean(bomb_planted %in% TRUE, na.rm = TRUE),
    mean_duration_s = mean(round_duration_s, na.rm = TRUE),
    median_duration_s = median(round_duration_s, na.rm = TRUE),
    ct_round_wins = sum(ct_win %in% TRUE, na.rm = TRUE),
    t_round_wins = sum(ct_win %in% FALSE, na.rm = TRUE),
    .groups = "drop"
  )
# Keep the PDF table compact; the complete table is still exported to CSV.
show_table(rounds_by_map |> select(
  map, matches("^(matches|rounds|pistol_rounds|overtime_rounds|planted_rounds|planted_pct)$"),
  ct_round_wins, t_round_wins
))
```

| map | matches | rounds | pistol_rounds | overtime_rounds | planted_rounds | planted_pct | ct_round_ |
|---|---|---|---|---|---|---|---|
| Dust2 | 30 | 628 | 60 | 14 | 349 | 55.57 | |
| Mirage | 30 | 629 | 60 | 16 | 312 | 49.60 | |

```
save_csv(rounds_by_map, "qmd_rounds_by_map.csv")
```

### 3.3 2.3 Round duration distribution

```
p_duration <- ggplot(rounds, aes(x = round_duration_s, fill = map)) +
  geom_histogram(bins = 30, alpha = .75, position = "identity") +
  facet_wrap(~map, ncol = 1) +
  labs(title = "Round-duration distribution", x = "Round duration (seconds)", y = "Rounds")
p_duration
```



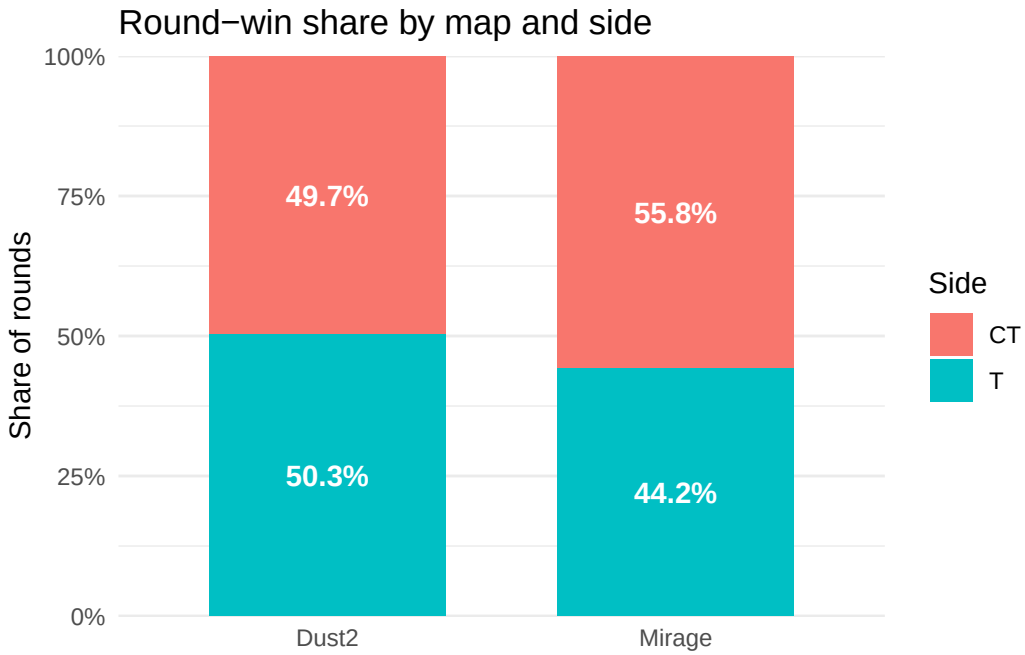

```
save_pdf(p_duration, "01_round_duration_distribution.pdf", 8, 7)
```

### 3.4 2.4 Round wins by side and map

Percentages are calculated within each map, so each map's stacked bar sums to 100%.

```
side_distribution <- rounds |>
  group_by(map) |>
  summarise(CT = mean(ct_win, na.rm = TRUE) * 100,
            T = mean(!ct_win, na.rm = TRUE) * 100, .groups = "drop") |>
  pivot_longer(c(CT, T), names_to = "side", values_to = "percentage")
p_side <- ggplot(side_distribution, aes(map, percentage, fill = side)) +
```

```
  geom_col(width = .68) +
  geom_text(aes(label = sprintf("%.1f%%", percentage)),
            position = position_stack(vjust = .5), colour = "white", fontface = "bold") +
  scale_y_continuous(limits = c(0, 100), labels = \(x) paste0(x, "%"),
                     expand = c(0, 0)) +
  labs(title = "Round-win share by map and side", x = NULL, y = "Share of rounds", fill = "Si
p_side
```




```
save_pdf(p_side, "02_side_win_share_by_map.pdf", 7, 5)
```

### 3.5 2.5 Win conditions by map

Missing `reason_name` values remain missing and are displayed as `NA`; they are not relabelled as a specific win condition.

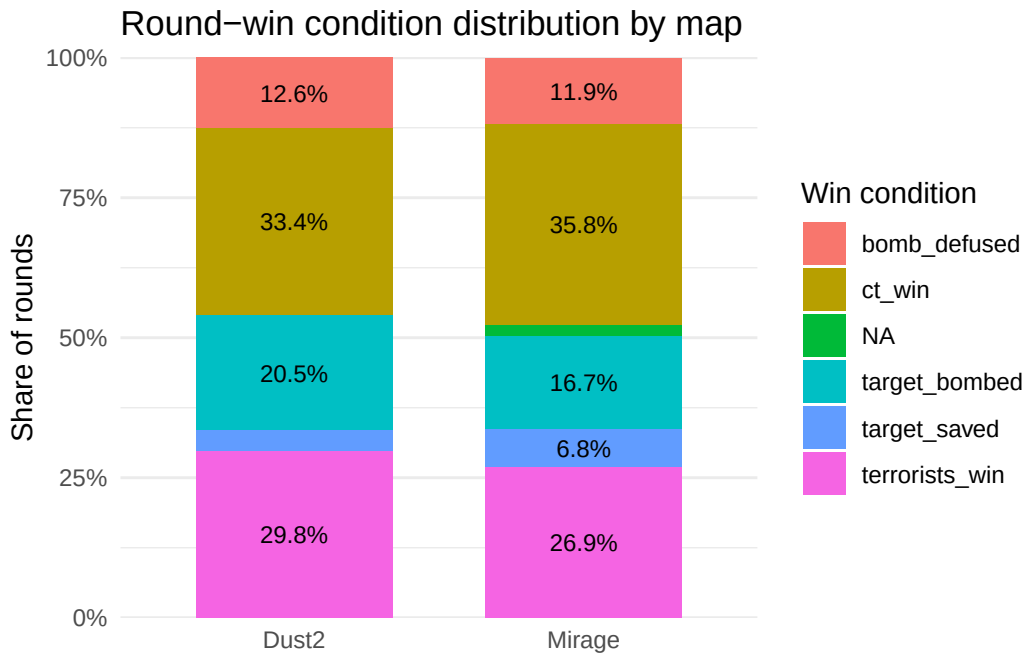
```
win_condition_summary <- rounds |>
  group_by(map, reason_name) |>
  summarise(rounds = n(), .groups = "drop_last") |>
  mutate(pct_of_map_rounds = 100 * rounds / sum(rounds)) |>
  ungroup() |>
```

```
  mutate(reason_label = if_else(is.na(reason_name), "NA", as.character(reason_name)))
show_table(win_condition_summary |> select(map, reason_label, rounds, pct_of_map_rounds))
```

| map | reason_label | rounds | pct_of_map_rounds |
|---|---|---|---|
| Dust2 | bomb_defused | 79 | 12.58 |
| Dust2 | ct_win | 210 | 33.44 |
| Dust2 | target_bombed | 129 | 20.54 |
| Dust2 | target_saved | 23 | 3.66 |
| Dust2 | terrorists_win | 187 | 29.78 |
| Mirage | bomb_defused | 75 | 11.92 |
| Mirage | ct_win | 225 | 35.77 |
| Mirage | target_bombed | 105 | 16.69 |
| Mirage | target_saved | 43 | 6.84 |
| Mirage | terrorists_win | 169 | 26.87 |
| Mirage | NA | 12 | 1.91 |

```
save_csv(win_condition_summary |> select(map, reason_label, rounds, pct_of_map_rounds), "qmd

p_conditions <- ggplot(win_condition_summary,
                       aes(x = map, y = pct_of_map_rounds, fill = reason_label)) +
  geom_col(width = .68) +
  geom_text(aes(label = if_else(pct_of_map_rounds >= 5,
                                sprintf("%.1f%%", pct_of_map_rounds), "")),
            position = position_stack(vjust = .5), size = 3) +
  scale_y_continuous(limits = c(0, 100), labels = \(x) paste0(x, "%"),
                     expand = c(0, 0)) +
  labs(title = "Round-win condition distribution by map", x = NULL,
       y = "Share of rounds", fill = "Win condition")
p_conditions
```




```
save_pdf(p_conditions, "03_win_condition_distribution.pdf", 9, 6)
```

### 3.6 2.6 Team performance by map and side

Each round contributes one observation for each participating team, assigned to its side for that round.

```
team_side_rounds <- bind_rows(
  rounds |> transmute(match_id, round_num, map, team = ct_team, side = "CT",
                      side_won = ct_win %in% TRUE, winner_team, round_duration_s,
                      is_pistol_round, is_overtime),
  rounds |> transmute(match_id, round_num, map, team = t_team, side = "T",
                      side_won = ct_win %in% FALSE, winner_team, round_duration_s,
                      is_pistol_round, is_overtime)
) |>
  filter(!is.na(team), team != "")
team_map_side_summary <- team_side_rounds |>
  group_by(map, team, side) |>
  summarise(
    matches = n_distinct(match_id), rounds_played = n(),
    rounds_won_on_side = sum(side_won, na.rm = TRUE),
    side_round_win_pct = 100 * mean(side_won, na.rm = TRUE),
```

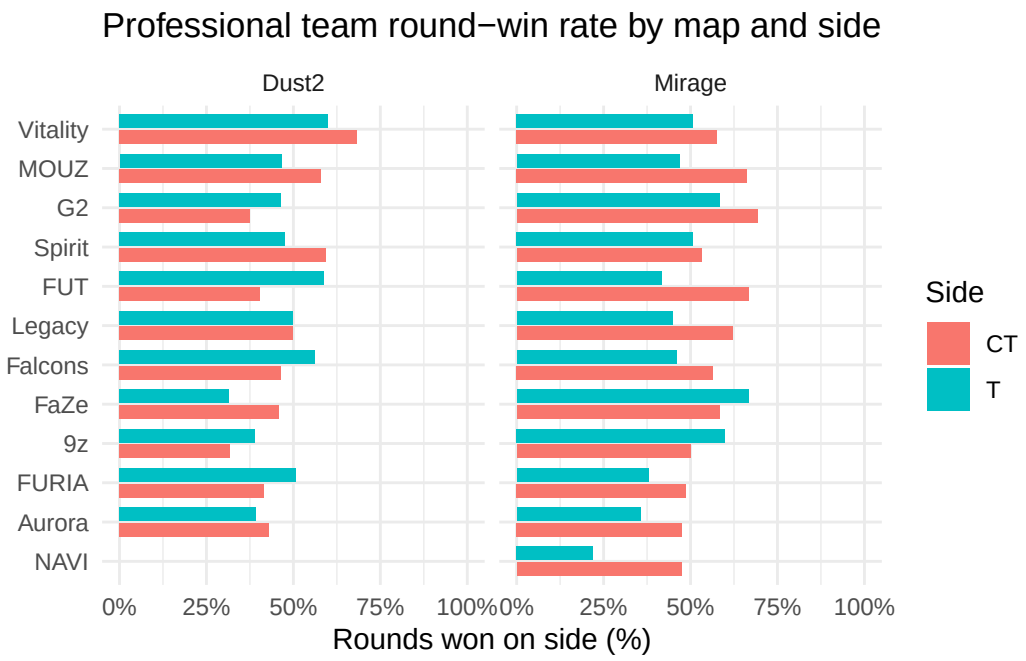
```
    mean_round_duration_s = mean(round_duration_s, na.rm = TRUE),
    pistol_rounds = sum(is_pistol_round %in% TRUE, na.rm = TRUE),
    overtime_rounds = sum(is_overtime %in% TRUE, na.rm = TRUE),
    .groups = "drop"
  )
show_table(team_map_side_summary |> arrange(map, team, side))
```

| map | team | side | matches | rounds_played | rounds_won_on_side | side_round_win_pct | mean_ |
|---|---|---|---|---|---|---|---|
| Dust2 | 9z | CT | 3 | 22 | 7 | 31.82 | |
| Dust2 | 9z | T | 3 | 36 | 14 | 38.89 | |
| Dust2 | Aurora | CT | 2 | 21 | 9 | 42.86 | |
| Dust2 | Aurora | T | 2 | 23 | 9 | 39.13 | |
| Dust2 | FURIA | CT | 8 | 82 | 34 | 41.46 | |
| Dust2 | FURIA | T | 8 | 77 | 39 | 50.65 | |
| Dust2 | FUT | CT | 4 | 52 | 21 | 40.38 | |
| Dust2 | FUT | T | 4 | 51 | 30 | 58.82 | |
| Dust2 | FaZe | CT | 2 | 24 | 11 | 45.83 | |
| Dust2 | FaZe | T | 2 | 19 | 6 | 31.58 | |
| Dust2 | Falcons | CT | 5 | 56 | 26 | 46.43 | |
| Dust2 | Falcons | T | 5 | 48 | 27 | 56.25 | |
| Dust2 | G2 | CT | 6 | 72 | 27 | 37.50 | |
| Dust2 | G2 | T | 6 | 43 | 20 | 46.51 | |
| Dust2 | Legacy | CT | 7 | 72 | 36 | 50.00 | |
| Dust2 | Legacy | T | 7 | 74 | 37 | 50.00 | |
| Dust2 | MOUZ | CT | 7 | 88 | 51 | 57.95 | |
| Dust2 | MOUZ | T | 7 | 73 | 34 | 46.58 | |
| Dust2 | Spirit | CT | 7 | 54 | 32 | 59.26 | |
| Dust2 | Spirit | T | 7 | 84 | 40 | 47.62 | |
| Dust2 | Vitality | CT | 9 | 85 | 58 | 68.24 | |
| Dust2 | Vitality | T | 9 | 100 | 60 | 60.00 | |
| Mirage | 9z | CT | 1 | 14 | 7 | 50.00 | |
| Mirage | 9z | T | 1 | 15 | 9 | 60.00 | |
| Mirage | Aurora | CT | 8 | 82 | 39 | 47.56 | |
| Mirage | Aurora | T | 8 | 73 | 26 | 35.62 | |
| Mirage | FURIA | CT | 7 | 82 | 40 | 48.78 | |
| Mirage | FURIA | T | 7 | 87 | 33 | 37.93 | |
| Mirage | FUT | CT | 1 | 12 | 8 | 66.67 | |
| Mirage | FUT | T | 1 | 12 | 5 | 41.67 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Mirage | FaZe | CT | 1 | 12 | 7 | 58.33 |
| Mirage | FaZe | T | 1 | 9 | 6 | 66.67 |
| Mirage | Falcons | CT | 6 | 69 | 39 | 56.52 |
| Mirage | Falcons | T | 6 | 50 | 23 | 46.00 |
| Mirage | G2 | CT | 2 | 13 | 9 | 69.23 |
| Mirage | G2 | T | 2 | 24 | 14 | 58.33 |
| Mirage | Legacy | CT | 6 | 58 | 36 | 62.07 |
| Mirage | Legacy | T | 6 | 67 | 30 | 44.78 |
| Mirage | MOUZ | CT | 9 | 92 | 61 | 66.30 |
| Mirage | MOUZ | T | 9 | 111 | 52 | 46.85 |
| Mirage | NAVI | CT | 4 | 38 | 18 | 47.37 |
| Mirage | NAVI | T | 4 | 41 | 9 | 21.95 |
| Mirage | Spirit | CT | 7 | 79 | 42 | 53.16 |
| Mirage | Spirit | T | 7 | 63 | 32 | 50.79 |
| Mirage | Vitality | CT | 8 | 78 | 45 | 57.69 |
| Mirage | Vitality | T | 8 | 77 | 39 | 50.65 |

```
save_csv(team_map_side_summary |> arrange(map, team, side), "qmd_team_map_side_summary.csv")

p_team_side <- ggplot(team_map_side_summary,
                      aes(x = reorder(team, side_round_win_pct),
                          y = side_round_win_pct, fill = side)) +
  geom_col(position = position_dodge(width = .8), width = .7) +
  coord_flip() + facet_wrap(~map) +
  scale_y_continuous(limits = c(0, 100), labels = \(x) paste0(x, "%")) +
  labs(title = "Professional team round-win rate by map and side",
       x = NULL, y = "Rounds won on side (%)", fill = "Side")
p_team_side
```




```
save_pdf(p_team_side, "04_team_win_rate_by_map_side.pdf", 12, 9)
```

### 3.7 2.7 Team round-win distribution by map

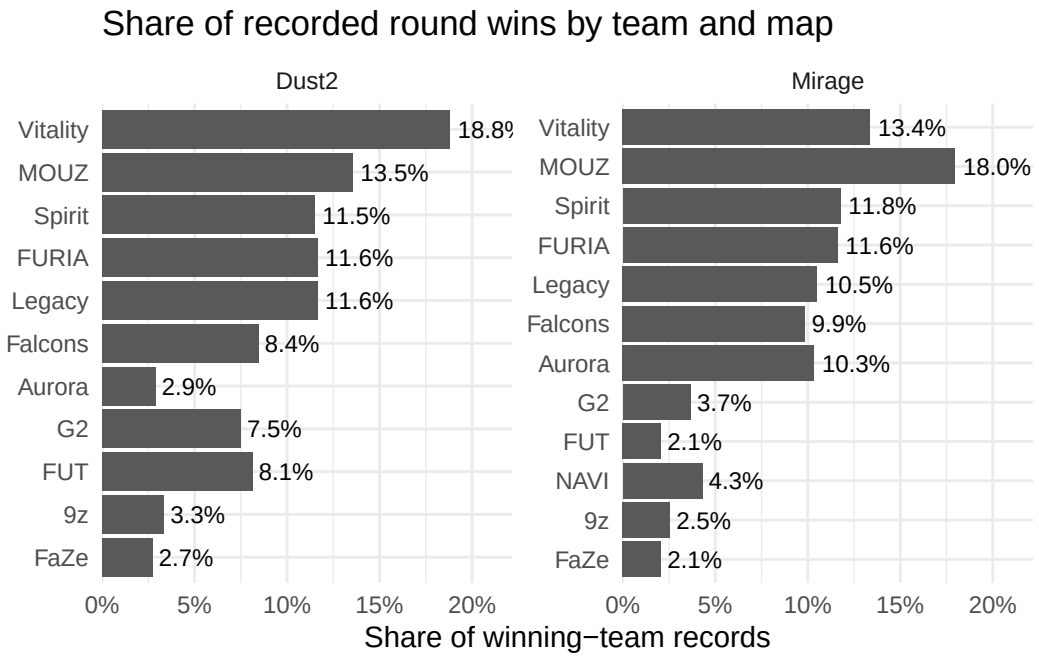
This is a count of round-level `winner_team` values, not a reconstructed match winner.

```
team_round_outcomes <- rounds |>
  filter(!is.na(winner_team), winner_team != "") |>
  count(map, winner_team, name = "rounds_won") |>
  group_by(map) |>
  mutate(pct_of_recorded_round_wins = 100 * rounds_won / sum(rounds_won)) |>
  ungroup() |>
  arrange(map, desc(rounds_won))
show_table(team_round_outcomes)
```

| map | winner_team | rounds_won | pct_of_recorded_round_wins |
|---|---|---|---|
| Dust2 | Vitality | 118 | 18.79 |
| Dust2 | MOUZ | 85 | 13.54 |
| Dust2 | FURIA | 73 | 11.62 |
| Dust2 | Legacy | 73 | 11.62 |
| Dust2 | Spirit | 72 | 11.46 |
| Dust2 | Falcons | 53 | 8.44 |
| Dust2 | FUT | 51 | 8.12 |
| Dust2 | G2 | 47 | 7.48 |
| Dust2 | 9z | 21 | 3.34 |
| Dust2 | Aurora | 18 | 2.87 |
| Dust2 | FaZe | 17 | 2.71 |
| Mirage | MOUZ | 113 | 17.97 |
| Mirage | Vitality | 84 | 13.35 |
| Mirage | Spirit | 74 | 11.76 |
| Mirage | FURIA | 73 | 11.61 |
| Mirage | Legacy | 66 | 10.49 |
| Mirage | Aurora | 65 | 10.33 |
| Mirage | Falcons | 62 | 9.86 |
| Mirage | NAVI | 27 | 4.29 |
| Mirage | G2 | 23 | 3.66 |
| Mirage | 9z | 16 | 2.54 |
| Mirage | FUT | 13 | 2.07 |
| Mirage | FaZe | 13 | 2.07 |

```
save_csv(team_round_outcomes, "qmd_team_round_outcomes.csv")

p_team_wins <- ggplot(team_round_outcomes,
                      aes(x = reorder(winner_team, rounds_won),
                          y = pct_of_recorded_round_wins)) +
  geom_col() + coord_flip() + facet_wrap(~map, scales = "free_y") +
  geom_text(aes(label = sprintf("%.1f%%", pct_of_recorded_round_wins)),
            hjust = -.12, size = 3) +
  scale_y_continuous(expand = expansion(mult = c(0, .18)),
                     labels = \(x) paste0(x, "%")) +
  labs(title = "Share of recorded round wins by team and map",
       x = NULL, y = "Share of winning-team records")
p_team_wins
```




```
save_pdf(p_team_wins, "05_team_round_win_share.pdf", 12, 9)
```

# 4 3. Round-state analysis (round_states)

## 4.1 3.1 Snapshot coverage and sampling

```
snapshots_per_round <- round_states |>
  count(match_id, map, round_num, name = "snapshots")
show_table(snapshots_per_round |> summarise(
  rounds = n(), min_snapshots = min(snapshots),
  q1_snapshots = quantile(snapshots, .25),
  median_snapshots = median(snapshots),
  mean_snapshots = mean(snapshots),
  q3_snapshots = quantile(snapshots, .75),
  max_snapshots = max(snapshots)
))
```

| rounds | min_snapshots | q1_snapshots | median_snapshots | mean_snapshots | q3_snapshots | max_snap |
|---|---|---|---|---|---|---|
| 1257 | 14 | 71 | 95 | 92.44 | 115 | |

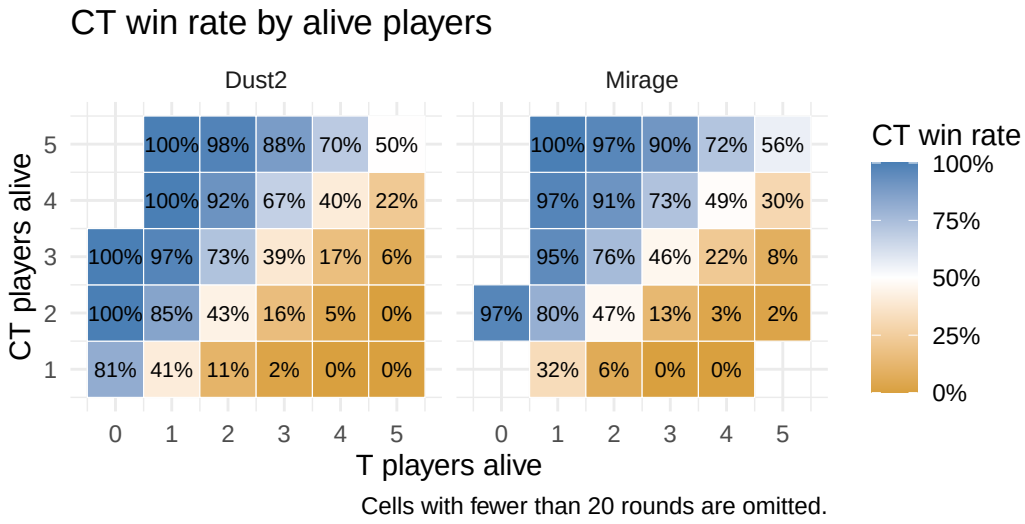
```r
save_csv(snapshots_per_round, "qmd_snapshots_per_round.csv")
show_table(round_states |> summarise(
  snapshots = n(), min_seconds = min(seconds, na.rm = TRUE),
  median_seconds = median(seconds, na.rm = TRUE),
  max_seconds = max(seconds, na.rm = TRUE)
))
```

| snapshots | min_seconds | median_seconds | max_seconds |
|---|---|---|---|
| 116201 | 0 | 46 | 155 |

### 4.2 3.2 Alive-player advantage heatmap

Each round is counted once per distinct alive-count combination it passed through, so repeated snapshots with the same alive counts do not multiply that combination within a round. Cells with fewer than 20 rounds are omitted.

```r
alive <- round_states |>
  distinct(map, match_id, round_num, ct_alive, t_alive, ct_win) |>
  group_by(map, ct_alive, t_alive) |>
  summarise(ct_win_rate = mean(ct_win, na.rm = TRUE), n = n(), .groups = "drop") |>
  filter(n >= 20)
p_alive <- ggplot(alive, aes(t_alive, ct_alive, fill = ct_win_rate)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.0f%%", 100 * ct_win_rate)), size = 2.8) +
  scale_fill_gradient2(low = "#d9a03f", mid = "white", high = "#4a7fb5",
                       midpoint = .5, limits = c(0, 1), labels = percent) +
  scale_x_continuous(breaks = 0:5) + scale_y_continuous(breaks = 0:5) +
  coord_equal() + facet_wrap(~map) +
  labs(title = "CT win rate by alive players", x = "T players alive",
       y = "CT players alive", fill = "CT win rate",
       caption = "Cells with fewer than 20 rounds are omitted.")
p_alive
```




```
save_pdf(p_alive, "06_alive_player_win_rate_heatmap.pdf", 8, 5)
```

### 4.3 3.3 Round-end alive players and health

For each round, the round-end snapshot is the latest recorded snapshot at or before `end_tick`. Snapshots are periodic, so this is an approximation and may not equal the exact end tick.

```
round_end <- round_states_enriched |>
  filter(!is.na(end_tick), tick <= end_tick) |>
  group_by(match_id, map, round_num) |>
  slice_max(tick, n = 1, with_ties = FALSE) |>
  ungroup()

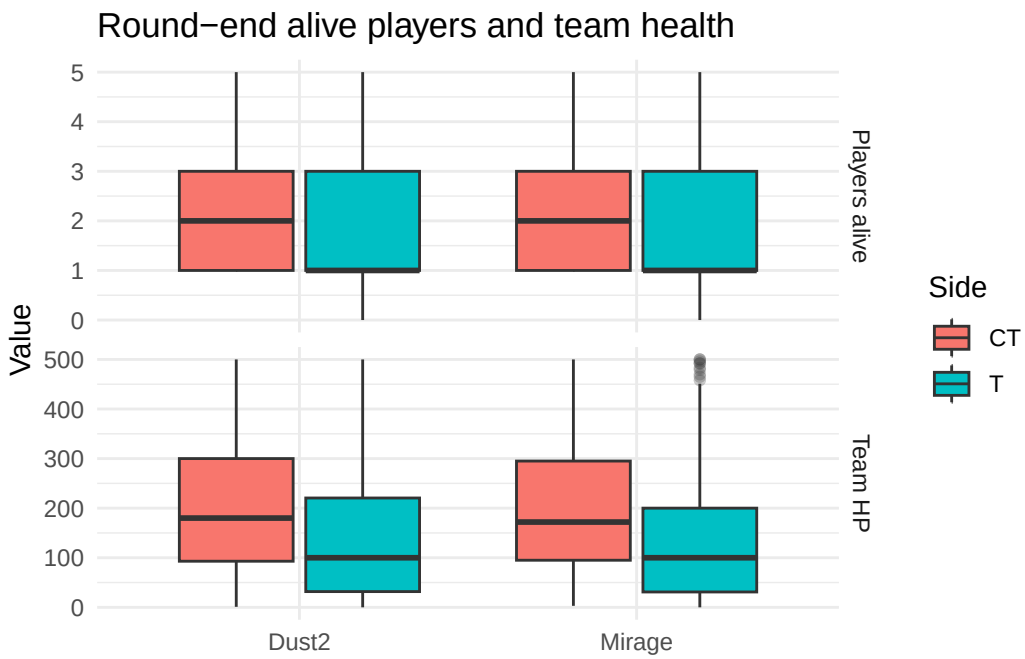
round_end_summary <- round_end |>
  group_by(map) |>
  summarise(
    rounds = n(),
    mean_ct_alive = mean(ct_alive, na.rm = TRUE),
    median_ct_alive = median(ct_alive, na.rm = TRUE),
    mean_t_alive = mean(t_alive, na.rm = TRUE),
    median_t_alive = median(t_alive, na.rm = TRUE),
    mean_ct_hp = mean(ct_hp, na.rm = TRUE),
```

```
    median_ct_hp = median(ct_hp, na.rm = TRUE),
    mean_t_hp = mean(t_hp, na.rm = TRUE),
    median_t_hp = median(t_hp, na.rm = TRUE),
    .groups = "drop"
  )
show_table(round_end_summary)
```

| map | rounds | mean_ct_alive | median_ct_alive | mean_t_alive | median_t_alive | mean_ct_hp | me |
|---|---|---|---|---|---|---|---|
| Dust2 | 628 | 2.33 | 2 | 1.91 | 1 | 190.80 | |
| Mirage | 629 | 2.35 | 2 | 1.76 | 1 | 189.99 | |

```
save_csv(round_end_summary, "qmd_round_end_summary.csv")

round_end_long <- round_end |>
  select(map, match_id, round_num, ct_alive, t_alive, ct_hp, t_hp) |>
  pivot_longer(c(ct_alive, t_alive, ct_hp, t_hp),
               names_to = "measure", values_to = "value") |>
  mutate(
    side = if_else(str_starts(measure, "ct_"), "CT", "T"),
    metric = if_else(str_ends(measure, "alive"), "Players alive", "Team HP")
  )
p_round_end <- ggplot(round_end_long, aes(map, value, fill = side)) +
  geom_boxplot(outlier.alpha = .25) +
  facet_grid(metric ~ ., scales = "free_y") +
  labs(title = "Round-end alive players and team health",
       x = NULL, y = "Value", fill = "Side")
p_round_end
```




```
save_pdf(p_round_end, "07_round_end_alive_and_health.pdf", 8, 7)
```

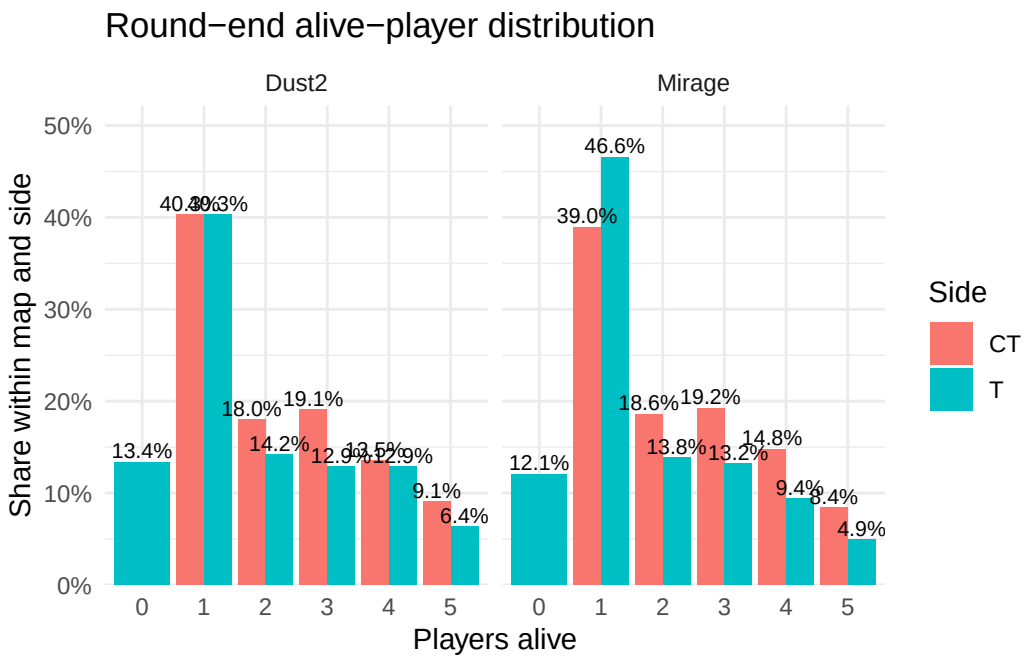
### 4.4 3.4 Round-end alive-count distribution

```
round_end_alive_dist <- round_end |>
  select(map, ct_alive, t_alive) |>
  pivot_longer(c(ct_alive, t_alive), names_to = "side", values_to = "alive") |>
  mutate(side = recode(side, ct_alive = "CT", t_alive = "T")) |>
  count(map, side, alive, name = "rounds") |>
  group_by(map, side) |>
  mutate(pct = 100 * rounds / sum(rounds)) |>
  ungroup()
show_table(round_end_alive_dist)
```

| map | side | alive | rounds | pct |
|---|---|---|---|---|
| Dust2 | CT | 1 | 253 | 40.29 |
| Dust2 | CT | 2 | 113 | 17.99 |
| Dust2 | CT | 3 | 120 | 19.11 |
| Dust2 | CT | 4 | 85 | 13.54 |
| Dust2 | CT | 5 | 57 | 9.08 |
| Dust2 | T | 0 | 84 | 13.38 |
| Dust2 | T | 1 | 253 | 40.29 |
| Dust2 | T | 2 | 89 | 14.17 |
| Dust2 | T | 3 | 81 | 12.90 |
| Dust2 | T | 4 | 81 | 12.90 |
| Dust2 | T | 5 | 40 | 6.37 |
| Mirage | CT | 1 | 245 | 38.95 |
| Mirage | CT | 2 | 117 | 18.60 |
| Mirage | CT | 3 | 121 | 19.24 |
| Mirage | CT | 4 | 93 | 14.79 |
| Mirage | CT | 5 | 53 | 8.43 |
| Mirage | T | 0 | 76 | 12.08 |
| Mirage | T | 1 | 293 | 46.58 |
| Mirage | T | 2 | 87 | 13.83 |
| Mirage | T | 3 | 83 | 13.20 |
| Mirage | T | 4 | 59 | 9.38 |
| Mirage | T | 5 | 31 | 4.93 |

```
save_csv(round_end_alive_dist, "qmd_round_end_alive_distribution.csv")

p_alive_dist <- ggplot(round_end_alive_dist,
                       aes(factor(alive, levels = 0:5), pct, fill = side)) +
  geom_col(position = "dodge") + facet_wrap(~map) +
  geom_text(aes(label = if_else(pct >= 4, sprintf("%.1f%%", pct), "")),
            position = position_dodge(width = .9), vjust = -.25, size = 2.8) +
  scale_y_continuous(labels = \(x) paste0(x, "%"),
                     expand = expansion(mult = c(0, .12))) +
  labs(title = "Round-end alive-player distribution",
       x = "Players alive", y = "Share within map and side", fill = "Side")
p_alive_dist
```




```
save_pdf(p_alive_dist, "08_round_end_alive_distribution.pdf", 9, 5)
```

# 5 4. Enriched round-state analysis (`round_states_enriched`)

## 5.1 4.1 Enriched data checks

The join should preserve the snapshot row count. Round-level fields repeat on every snapshot belonging to that round; therefore, missing values in fields copied from `rounds` are repeated across all snapshots for the affected round. This repetition is expected and explains why missing-cell counts can be much larger in the enriched table than in the one-row-per-round source.

```
enriched_checks <- tibble(
  dataset = c("round_states", "round_states_enriched"),
  rows = c(nrow(round_states), nrow(round_states_enriched)),
  columns = c(ncol(round_states), ncol(round_states_enriched)),
  distinct_round_keys = c(
    nrow(distinct(round_states, match_id, map, round_num)),
    nrow(distinct(round_states_enriched, match_id, map, round_num))
  )
```

```
)
show_table(enriched_checks)
```

| dataset | rows | columns | distinct_round_keys |
|---|---|---|---|
| round_states | 116201 | 45 | 1257 |
| round_states_enriched | 116201 | 59 | 1257 |

```
save_csv(enriched_checks, "qmd_enriched_data_checks.csv")

# Count distinct round keys present in the enriched table but absent from rounds.
round_key_cols <- c("match_id", "map", "round_num")
round_keys <- rounds |> distinct(across(all_of(round_key_cols)))
enriched_keys <- round_states_enriched |> distinct(across(all_of(round_key_cols)))
enriched_rounds_not_in_rounds <- anti_join(enriched_keys, round_keys, by = round_key_cols) |>
  nrow()
join_validation <- tibble(
  check = "enriched_rounds_not_in_rounds",
  value = enriched_rounds_not_in_rounds,
  interpretation = if_else(
    enriched_rounds_not_in_rounds == 0,
    "All enriched round keys match a round in rounds; no unmatched enriched rounds.",
    "Some enriched round keys have no matching round in rounds; inspect the join keys."
  )
)
show_table(join_validation)
```

| check | value | interpretation |
|---|---|---|
| enriched_rounds_not_in_rounds | 0 | All enriched round keys match a round in rounds; no unmatched |

```
save_csv(join_validation, "qmd_enriched_round_join_validation.csv")
```

## 5.2 Within-round consistency check

This check counts how many rounds contain more than one distinct non-missing value for each selected field across their snapshots. It is intended to identify fields that should remain constant within a round but unexpectedly vary. `bomb_planted` is the exception: it is a time-varying state, normally `FALSE` before the bomb is planted and `TRUE` afterward. Therefore, the 626 rounds flagged for `bomb_planted` are not automatically errors; they indicate rounds where the snapshots record both states. The other listed fields have zero rounds with multiple values, which is consistent with them remaining constant within each round.

```r
constant_check_fields <- intersect(
  c("winner_side", "winner_team", "reason_name", "is_overtime",
    "is_pistol_round", "bomb_planted", "ct_buy_type", "t_buy_type",
    "first_death_side", "round_duration_s"),
  names(round_states_enriched)
)
round_constant_checks <- map_dfr(constant_check_fields, function(field_name) {
  per_round <- round_states_enriched |>
    group_by(match_id, map, round_num) |>
    summarise(
      distinct_nonmissing_values = n_distinct(.data[[field_name]][!is.na(.data[[field_name]])]),
      .groups = "drop"
    )
  tibble(
    field = field_name,
    rounds_with_multiple_values = sum(per_round$distinct_nonmissing_values > 1)
  )
}) |>
  mutate(
    interpretation = if_else(
      field == "bomb_planted",
      "Time-varying field: FALSE before plant and TRUE after plant is expected",
      "Expected to remain constant within a round; non-zero counts merit investigation"
    )
  )
show_table(round_constant_checks)
```

| field | rounds_with_multiple_values | interpretation |
|---|---|---|
| winner_side | 0 | Expected to remain constant within a round; non-zero |
| winner_team | 0 | Expected to remain constant within a round; non-zero |
| reason_name | 0 | Expected to remain constant within a round; non-zero |
| is_overtime | 0 | Expected to remain constant within a round; non-zero |
| is_pistol_round | 0 | Expected to remain constant within a round; non-zero |
| bomb_planted | 626 | Time-varying field: FALSE before plant and TRUE aft |
| ct_buy_type | 0 | Expected to remain constant within a round; non-zero |
| t_buy_type | 0 | Expected to remain constant within a round; non-zero |
| first_death_side | 0 | Expected to remain constant within a round; non-zero |
| round_duration_s | 0 | Expected to remain constant within a round; non-zero |

```r
save_csv(round_constant_checks, "qmd_round_constant_checks.csv")
```

```r
missing_enriched <- missing_by_dataset(round_states_enriched, "round_states_enriched")
show_table(missing_enriched)
```

| dataset | column | missing_n | missing_pct |
|---|---|---|---|
| round_states_enriched | seconds_since_plant | 97855 | 84.21 |
| round_states_enriched | plant_site | 97855 | 84.21 |
| round_states_enriched | plant_tick | 47215 | 40.63 |
| round_states_enriched | ct_won_previous_round | 4116 | 3.54 |
| round_states_enriched | reason_name | 1263 | 1.09 |
| round_states_enriched | ct_buy_type | 1198 | 1.03 |
| round_states_enriched | t_buy_type | 1198 | 1.03 |
| round_states_enriched | first_death_side | 217 | 0.19 |

```r
save_csv(missing_enriched, "qmd_missing_values_enriched.csv")
```

### 5.3 4.2 Bomb plant rates and plant-site distribution

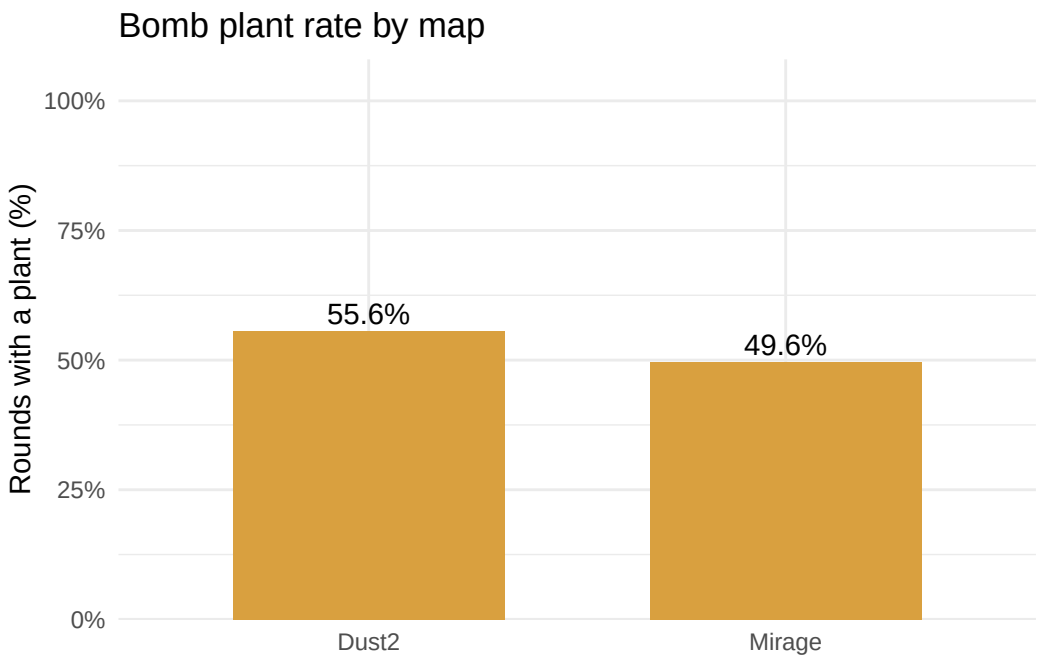
```r
plant_summary <- rounds |>
  group_by(map) |>
  summarise(
    rounds = n(),
    planted_rounds = sum(bomb_planted %in% TRUE, na.rm = TRUE),
    plant_rate_pct = 100 * mean(bomb_planted %in% TRUE, na.rm = TRUE),
    .groups = "drop"
  )
show_table(plant_summary)
```

| map | rounds | planted_rounds | plant_rate_pct |
|---|---|---|---|
| Dust2 | 628 | 349 | 55.57 |
| Mirage | 629 | 312 | 49.60 |

```r
save_csv(plant_summary, "qmd_bomb_plant_summary.csv")

p_plant <- ggplot(plant_summary, aes(map, plant_rate_pct)) +
  geom_col(width = .65, fill = "#d9a03f") +
  geom_text(aes(label = sprintf("%.1f%%", plant_rate_pct)), vjust = -.35) +
  scale_y_continuous(limits = c(0, 100), labels = \(x) paste0(x, "%"),
                     expand = expansion(mult = c(0, .08))) +
```

```
  labs(title = "Bomb plant rate by map", x = NULL, y = "Rounds with a plant (%)")
p_plant
```



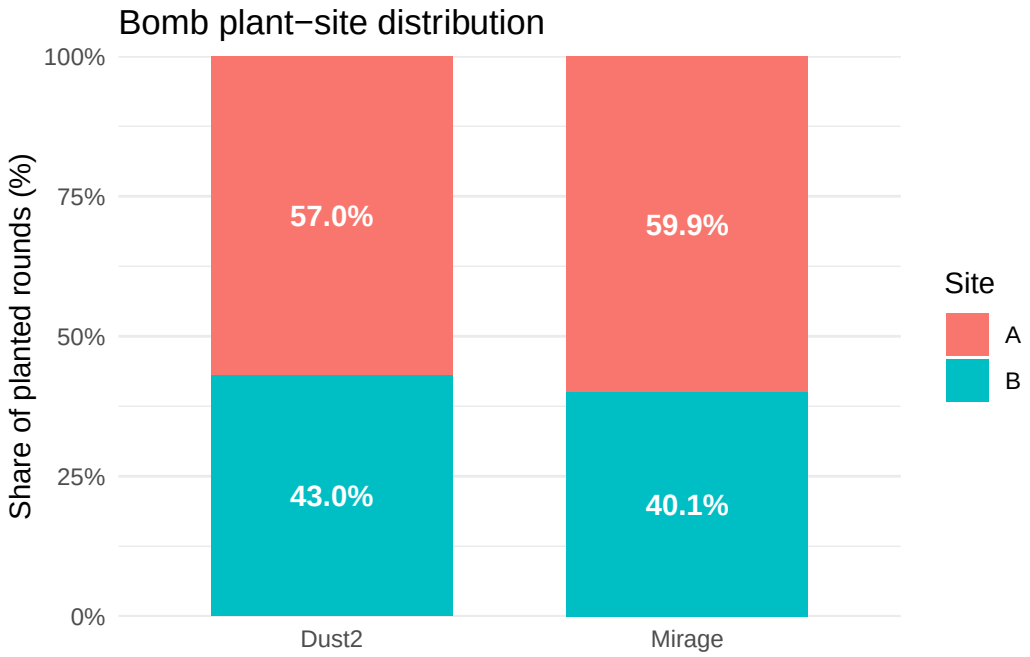

```
save_pdf(p_plant, "09_bomb_plant_rate.pdf", 7, 5)

plant_site <- rounds |>
  filter(bomb_planted %in% TRUE, !is.na(plant_site)) |>
  count(map, plant_site, name = "plants") |>
  group_by(map) |>
  mutate(pct = 100 * plants / sum(plants)) |>
  ungroup()
show_table(plant_site)
```

| map | plant_site | plants | pct |
|---|---|---|---|
| Dust2 | A | 199 | 57.02 |
| Dust2 | B | 150 | 42.98 |
| Mirage | A | 187 | 59.94 |
| Mirage | B | 125 | 40.06 |

```
save_csv(plant_site, "qmd_plant_site_distribution.csv")
```

```r
p_site <- ggplot(plant_site, aes(map, pct, fill = as.character(plant_site))) +
  geom_col(width = .68) +
  geom_text(aes(label = sprintf("%.1f%%", pct)),
            position = position_stack(vjust = .5), colour = "white", fontface = "bold") +
  scale_y_continuous(limits = c(0, 100), labels = \(x) paste0(x, "%"),
                     expand = c(0, 0)) +
  labs(title = "Bomb plant-site distribution", x = NULL,
       y = "Share of planted rounds (%)", fill = "Site")
p_site
```




```r
save_pdf(p_site, "10_bomb_plant_site_distribution.pdf", 7, 5)
```

## 5.4 4.3 T win rate with and without a plant

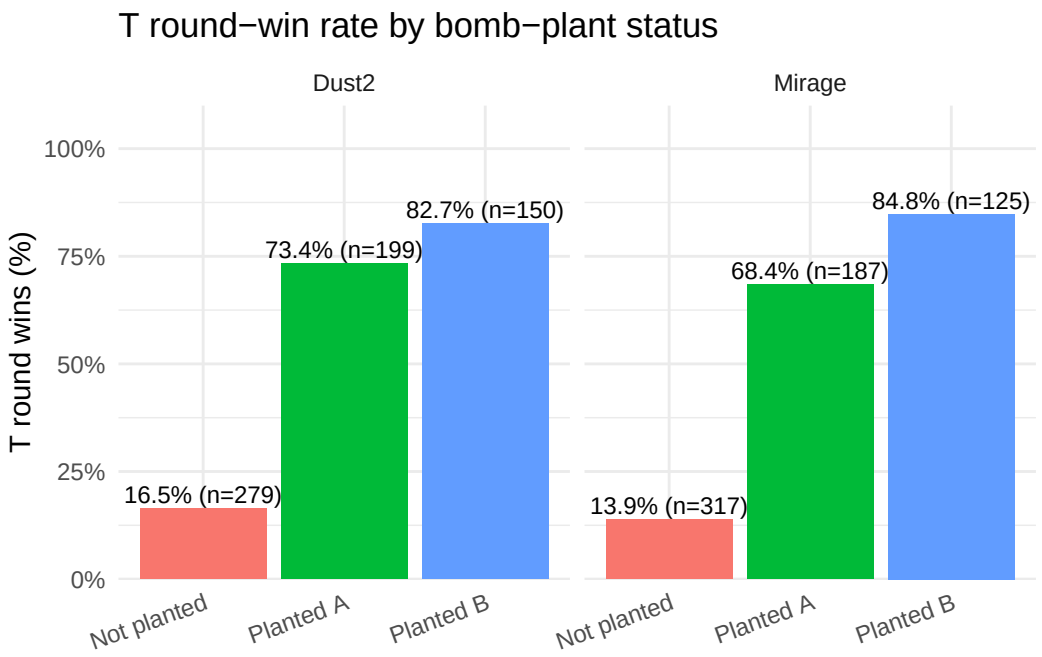
```r
plant_outcomes <- rounds |>
  mutate(plant_status = case_when(
    bomb_planted %in% TRUE ~ paste("Planted", coalesce(as.character(plant_site), "unknown si
    bomb_planted %in% FALSE ~ "Not planted",
    TRUE ~ "Unknown"
  )) |>
```

```
  group_by(map, plant_status) |>
  summarise(
    rounds = n(),
    t_win_pct = 100 * mean(!ct_win, na.rm = TRUE),
    .groups = "drop"
  )
show_table(plant_outcomes)
```

| map | plant_status | rounds | t_win_pct |
|---|---|---|---|
| Dust2 | Not planted | 279 | 16.49 |
| Dust2 | Planted A | 199 | 73.37 |
| Dust2 | Planted B | 150 | 82.67 |
| Mirage | Not planted | 317 | 13.88 |
| Mirage | Planted A | 187 | 68.45 |
| Mirage | Planted B | 125 | 84.80 |

```
save_csv(plant_outcomes, "qmd_plant_outcomes.csv")

p_plant_outcome <- ggplot(plant_outcomes,
                          aes(plant_status, t_win_pct, fill = plant_status)) +
  geom_col(show.legend = FALSE) + facet_wrap(~map) +
  geom_text(aes(label = sprintf("%.1f%% (n=%d)", t_win_pct, rounds)),
            vjust = -.3, size = 3) +
  scale_y_continuous(limits = c(0, 100), labels = \(x) paste0(x, "%"),
                     expand = expansion(mult = c(0, .1))) +
  labs(title = "T round-win rate by bomb-plant status", x = NULL, y = "T round wins (%)") +
  theme(axis.text.x = element_text(angle = 20, hjust = 1))
p_plant_outcome
```




```
save_pdf(p_plant_outcome, "11_t_win_rate_by_plant_status.pdf", 9, 5)
```

## 5.5 4.4 Time until plant and post-plant duration

The first planted snapshot approximates the time of plant; the latest snapshot at or before `end_tick` approximates the final post-plant state. The elapsed time values are snapshot-based estimates.

```
plant_timing <- round_states_enriched |>
  filter(bomb_planted %in% TRUE) |>
  group_by(match_id, map, round_num) |>
  slice_min(tick, n = 1, with_ties = FALSE) |>
  ungroup() |>
  select(match_id, map, round_num, seconds)

post_plant_timing <- round_states_enriched |>
  filter(bomb_planted %in% TRUE, !is.na(end_tick), tick <= end_tick) |>
  group_by(match_id, map, round_num) |>
  slice_max(tick, n = 1, with_ties = FALSE) |>
  ungroup() |>
  select(match_id, map, round_num, seconds_since_plant, reason_name)
```

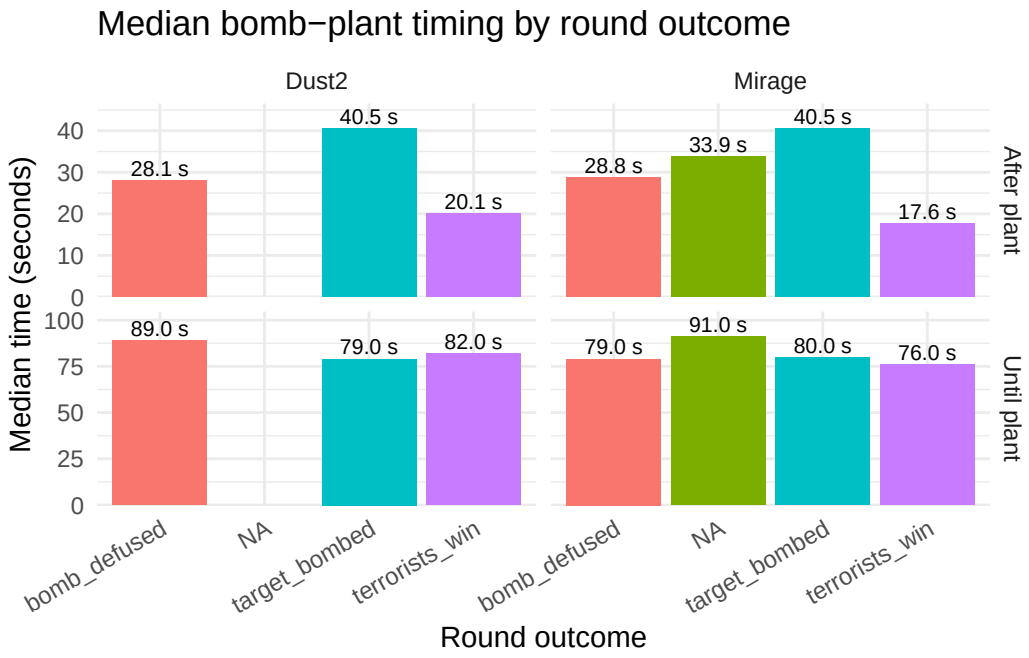
```
plant_timing_summary <- inner_join(plant_timing, post_plant_timing,
                                   by = c("match_id", "map", "round_num")) |>
  group_by(map, reason_name) |>
  summarise(
    planted_rounds = n(),
    median_seconds_until_plant = median(seconds, na.rm = TRUE),
    median_seconds_after_plant = median(seconds_since_plant, na.rm = TRUE),
    .groups = "drop"
  ) |>
  mutate(reason_label = if_else(is.na(reason_name), "NA", as.character(reason_name)))
show_table(plant_timing_summary |> select(map, reason_label, planted_rounds,
                                          median_seconds_until_plant,
                                          median_seconds_after_plant))
```

| map | reason_label | planted_rounds | median_seconds_until_plant | median_seconds_after_plant |
|---|---|---|---|---|
| Dust2 | bomb_defused | 79 | 89 | 28.08 |
| Dust2 | target_bombed | 129 | 79 | 40.48 |
| Dust2 | terrorists_win | 124 | 82 | 20.09 |
| Mirage | bomb_defused | 75 | 79 | 28.83 |
| Mirage | target_bombed | 105 | 80 | 40.48 |
| Mirage | terrorists_win | 107 | 76 | 17.64 |
| Mirage | NA | 7 | 91 | 33.86 |

```
save_csv(plant_timing_summary |> select(map, reason_label, planted_rounds,
                                          median_seconds_until_plant,
                                          median_seconds_after_plant), "qmd_plant_timing_summ

timing_long <- plant_timing_summary |>
  select(map, reason_label, median_seconds_until_plant, median_seconds_after_plant) |>
  pivot_longer(starts_with("median_seconds"),
               names_to = "timing", values_to = "median_seconds") |>
  mutate(timing = recode(timing,
    median_seconds_until_plant = "Until plant",
    median_seconds_after_plant = "After plant"))
p_timing <- ggplot(timing_long, aes(reason_label, median_seconds, fill = reason_label)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = sprintf("%.1f s", median_seconds)), vjust = -.3, size = 2.8) +
  facet_grid(timing ~ map, scales = "free_y") +
  scale_y_continuous(expand = expansion(mult = c(0, .15))) +
  labs(title = "Median bomb-plant timing by round outcome",
       x = "Round outcome", y = "Median time (seconds)") +
```

```
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
p_timing
```



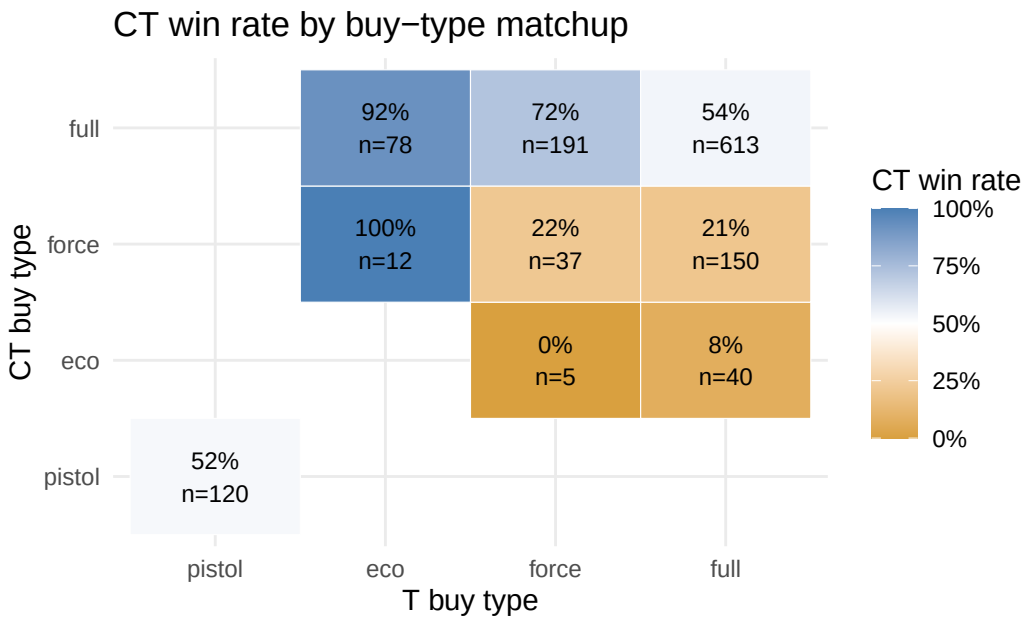

```
save_pdf(p_timing, "12_bomb_plant_timing.pdf", 11, 7)
```

### 5.6 4.5 CT win rate by buy-type matchup

```
buy_levels <- c("pistol", "eco", "force", "full")
buy_levels <- c(buy_levels, setdiff(unique(c(rounds$ct_buy_type, rounds$t_buy_type)),
                                    c(buy_levels, NA)))
buy <- rounds |>
  filter(!is.na(ct_buy_type), !is.na(t_buy_type)) |>
  mutate(across(c(ct_buy_type, t_buy_type), \(x) factor(x, levels = buy_levels))) |>
  group_by(ct_buy_type, t_buy_type) |>
  summarise(ct_win_rate = mean(ct_win, na.rm = TRUE), n = n(), .groups = "drop")
show_table(buy)
```

| ct_buy_type | t_buy_type | ct_win_rate | n |
|---|---|---|---|
| pistol | pistol | 0.52 | 120 |
| eco | force | 0.00 | 5 |
| eco | full | 0.07 | 40 |
| force | eco | 1.00 | 12 |
| force | force | 0.22 | 37 |
| force | full | 0.21 | 150 |
| full | eco | 0.92 | 78 |
| full | force | 0.72 | 191 |
| full | full | 0.54 | 613 |

```
save_csv(buy, "qmd_buy_type_matchup.csv")
p_buy <- ggplot(buy, aes(t_buy_type, ct_buy_type, fill = ct_win_rate)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.0f%%\nn=%d", 100 * ct_win_rate, n)), size = 3) +
  scale_fill_gradient2(low = "#d9a03f", mid = "white", high = "#4a7fb5",
                       midpoint = .5, limits = c(0, 1), labels = percent) +
  labs(title = "CT win rate by buy-type matchup", x = "T buy type",
       y = "CT buy type", fill = "CT win rate",
       caption = "Buy types are recorded at freeze end; missing buy types are excluded.")
p_buy
```




Buy types are recorded at freeze end; missing buy types are excluded.

```
save_pdf(p_buy, "13_buy_type_matchup_heatmap.pdf", 8, 6)
```

## 5.7 4.6 Round outcome versus alive-player advantage at 30 seconds

The selected row is the latest available snapshot at or before 30 seconds, not necessarily exactly at 30 seconds. One snapshot per round is used to avoid counting a round multiple times.

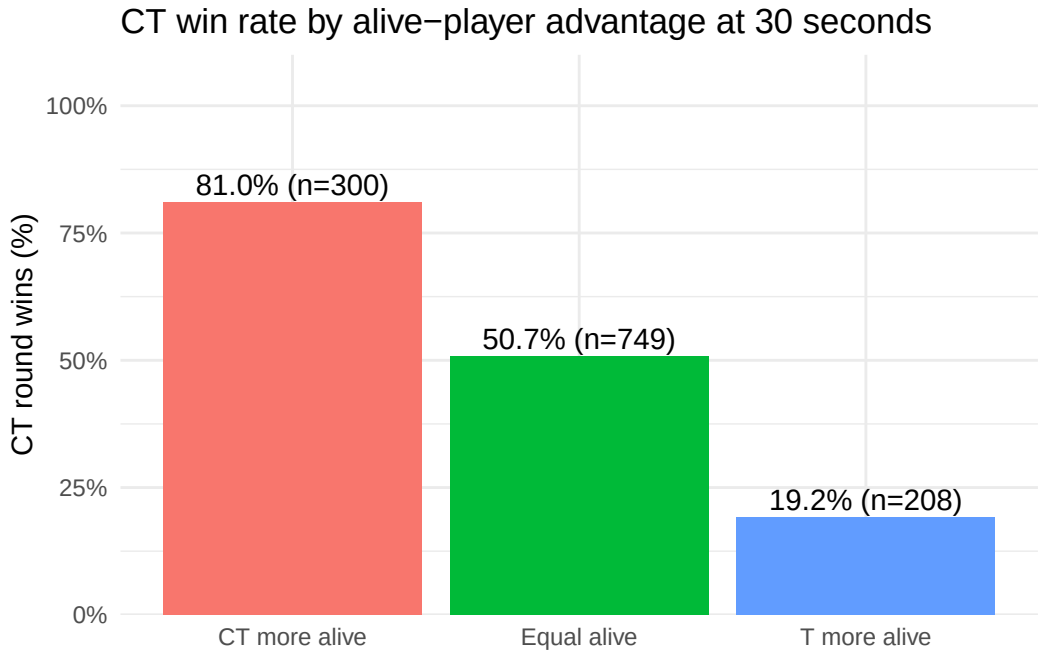
```
state_at_30s <- round_states_enriched |>
  filter(seconds <= 30) |>
  group_by(match_id, map, round_num) |>
  slice_max(seconds, n = 1, with_ties = FALSE) |>
  ungroup() |>
  mutate(alive_difference = ct_alive - t_alive,
         alive_advantage = case_when(
           alive_difference > 0 ~ "CT more alive",
           alive_difference < 0 ~ "T more alive",
           alive_difference == 0 ~ "Equal alive",
           TRUE ~ NA_character_
         ))
alive_outcome_30s <- state_at_30s |>
  filter(!is.na(ct_win), !is.na(alive_advantage)) |>
  group_by(alive_advantage) |>
  summarise(rounds = n(), ct_win_pct = 100 * mean(ct_win, na.rm = TRUE),
            .groups = "drop")
show_table(alive_outcome_30s)
```

| alive_advantage | rounds | ct_win_pct |
|---|---|---|
| CT more alive | 300 | 81.00 |
| Equal alive | 749 | 50.73 |
| T more alive | 208 | 19.23 |

```
save_csv(alive_outcome_30s, "qmd_alive_advantage_at_30s.csv")

p_alive_30 <- ggplot(alive_outcome_30s,
                     aes(alive_advantage, ct_win_pct, fill = alive_advantage)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = sprintf("%.1f%% (n=%d)", ct_win_pct, rounds)),
            vjust = -.35) +
  scale_y_continuous(limits = c(0, 100), labels = \(x) paste0(x, "%"),
                     expand = expansion(mult = c(0, .1))) +
```

```
  labs(title = "CT win rate by alive-player advantage at 30 seconds",
       x = NULL, y = "CT round wins (%)")
p_alive_30
```




```
save_pdf(p_alive_30, "14_alive_advantage_at_30s.pdf", 8, 5)
```

### 5.8 4.7 Equipment, money, health, and armor advantage at 30 seconds

```
state_economy_30 <- round_states_enriched |>
  filter(seconds <= 30) |>
  group_by(match_id, map, round_num) |>
  slice_max(seconds, n = 1, with_ties = FALSE) |>
  ungroup() |>
  mutate(
    equipment_difference = ct_equipment - t_equipment,
    money_difference = ct_money - t_money,
    hp_difference = ct_hp - t_hp,
    armor_difference = ct_armor - t_armor,
    equipment_advantage = case_when(
      equipment_difference > 0 ~ "CT higher equipment",
```

```
      equipment_difference < 0 ~ "T higher equipment",
      equipment_difference == 0 ~ "Equal equipment",
      TRUE ~ NA_character_
    ),
    hp_advantage = case_when(
      hp_difference > 0 ~ "CT higher HP",
      hp_difference < 0 ~ "T higher HP",
      hp_difference == 0 ~ "Equal HP",
      TRUE ~ NA_character_
    )
  )
economy_summary <- state_economy_30 |>
  filter(!is.na(ct_win), !is.na(equipment_advantage)) |>
  group_by(equipment_advantage) |>
  summarise(rounds = n(), ct_win_pct = 100 * mean(ct_win, na.rm = TRUE),
            mean_equipment_difference = mean(equipment_difference, na.rm = TRUE),
            mean_money_difference = mean(money_difference, na.rm = TRUE),
            .groups = "drop")
show_table(economy_summary)
```

| equipment_advantage | rounds | ct_win_pct | mean_equipment_difference | mean_money_difference |
|---|---|---|---|---|
| CT higher equipment | 793 | 68.73 | 9770.11 | 5265.07 |
| Equal equipment | 6 | 50.00 | 0.00 | -616.67 |
| T higher equipment | 458 | 25.11 | -8162.12 | -5150.87 |

```
save_csv(economy_summary, "qmd_equipment_advantage_summary.csv")

health_armor_summary <- state_economy_30 |>
  filter(!is.na(ct_win), !is.na(hp_advantage)) |>
  group_by(hp_advantage) |>
  summarise(rounds = n(), ct_win_pct = 100 * mean(ct_win, na.rm = TRUE),
            mean_hp_difference = mean(hp_difference, na.rm = TRUE),
            mean_armor_difference = mean(armor_difference, na.rm = TRUE),
            .groups = "drop")
show_table(health_armor_summary)
```

| hp_advantage | rounds | ct_win_pct | mean_hp_difference | mean_armor_difference |
|---|---|---|---|---|
| CT higher HP | 551 | 68.42 | 95.50 | 120.13 |
| Equal HP | 342 | 51.75 | 0.00 | 27.62 |
| T higher HP | 364 | 29.95 | -76.66 | -84.38 |

```r
save_csv(health_armor_summary, "qmd_health_armor_advantage_summary.csv")
```

# 6 5. Interpretation notes and limitations

- `rounds` is the round-level source; `round_states` contains multiple snapshots per round; the enriched table combines them for analysis while retaining snapshot granularity.
- Missing planting fields can be expected when no plant occurred. In the enriched data, any missing round-level value is repeated across every snapshot for that round, which can increase the missing-cell count.
- Percentages in stacked charts are calculated within their stated grouping (usually within map); sample sizes should be considered when interpreting percentages.
- Snapshot rows from the same round are not independent observations. Analyses intended to compare round outcomes should use one selected snapshot per round, as in the 30-second and round-end sections.
- `ct_win` is the eventual round outcome repeated across snapshots; do not treat every snapshot as an independent outcome for predictive modelling. If building predictive models, split training and test data by match or round to avoid leakage.
- Round-end and bomb timing analyses use periodic snapshots and therefore approximate the exact event times.
- All figures are saved as PDFs in `../report/figures/cs2-combined_analysis`, and CSV tables are saved in `../data/analysis/cs2-combined_analysis`.